

Source of Fields

STATIONARY CHARGES → ELECTROSTATIC
FIELDS

STEADY CURRENTS → MAGNETOSTATIC
FIELDS

TIME-VARYING CURRENTS

→ ELECTROMAGNETIC FIELDS, WAVES,
RADIATION

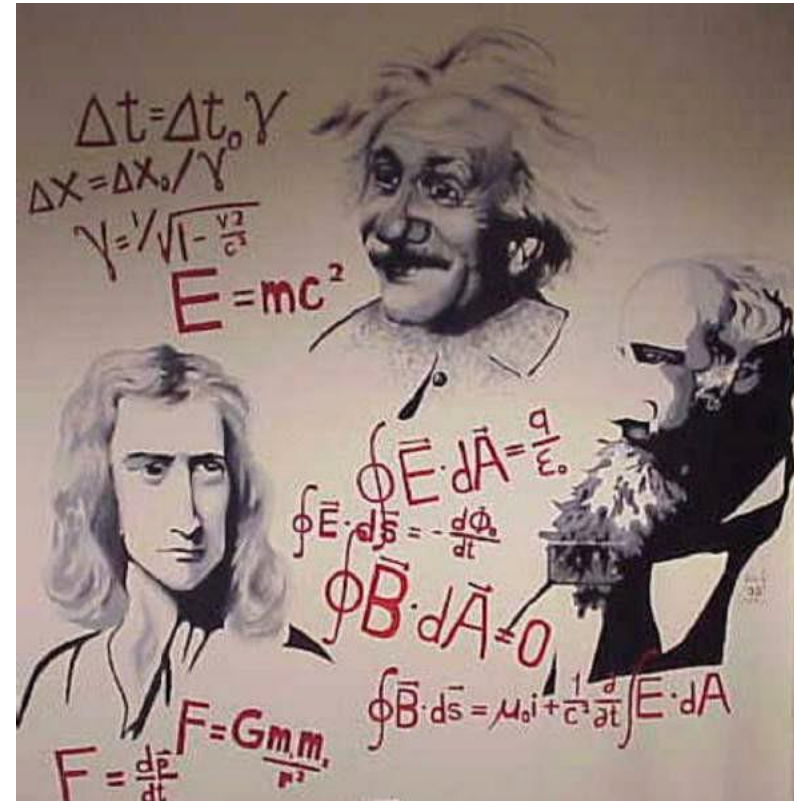
Who's Afraid of Maxwell's Equations?

“From a long view of the history of mankind - seen from, say, ten thousand years from now - there can be little doubt that the most significant event of the 19th century will be judged as Maxwell's discovery of the laws of electrodynamics”

(Richard P. Feynman)

The special theory of relativity owes its origins to Maxwell's equations of the electromagnetic field

(Albert Einstein)



“Maxwell can be justifiably placed with Einstein and Newton in a triad of the greatest physicists known to history”

(Ivan Tolstoy, Biographer)

Maxwell's equations

The greatest equations ever

- Although Maxwell's equations are relatively simple, they daringly reorganize our perception of nature, unifying electricity and magnetism and linking geometry, topology and physics
- They are essential to understanding the surrounding world and as the first field equations, they not only showed scientists a new way of approaching physics but also took them on the first step towards a **unification** of the fundamental forces of nature

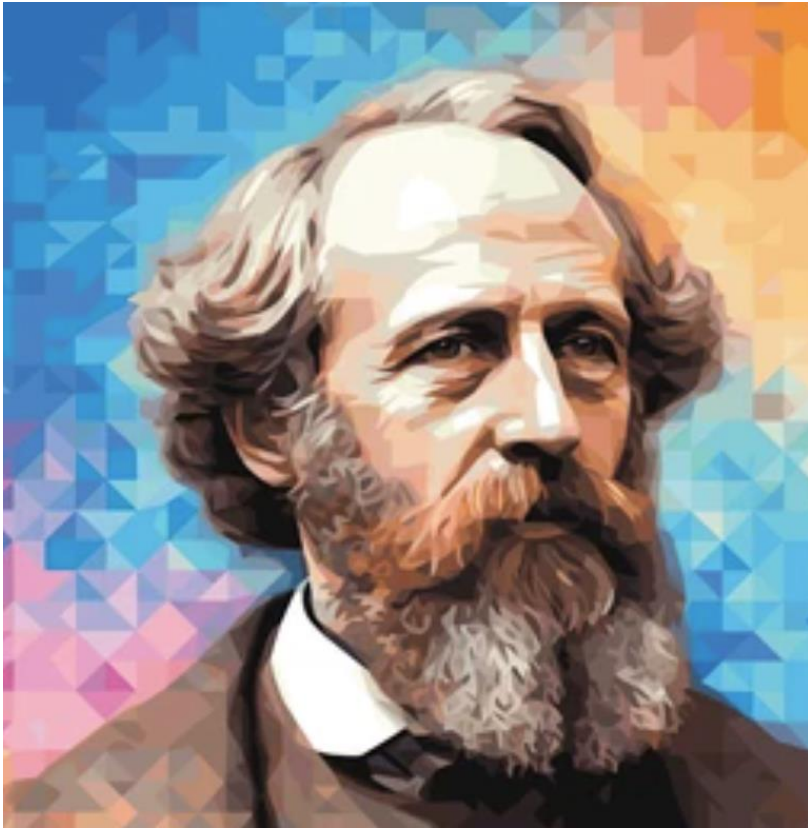
Unification

- **Gauss's Law for Electricity**: Gauss's law relates the electric field to the charge distribution within a closed surface.
 - Maxwell extended this law by introducing **the concept of electric flux and the divergence of the electric field**, thus providing a mathematical framework to describe how electric charges generate electric fields.
- **Ampère's Circuital Law**: Ampère's law deals with the relationship between the magnetic field, the closed loops of electric current, and the magnetic poles.
 - Maxwell extended Ampère's law by introducing the concept of **displacement current**. He recognized that a changing electric field could induce a magnetic field, similar to how Ampère's law describes the magnetic field generated by electric currents. This led to the addition of the displacement current term in the amended Ampère's law.

Unification

- **Faraday's Law of Electromagnetic Induction**: Faraday's law describes how a changing magnetic field induces an electromotive force (EMF) in a conductor.
 - Maxwell expanded on Faraday's law by **formalizing the mathematical equations** that govern electromagnetic induction. He also incorporated Faraday's work into his larger framework of equations, showing that changing magnetic fields could create electric fields.

Maxwell's Equations



$$\begin{aligned}\nabla \cdot \vec{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} &= \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)\end{aligned}$$

- Describe how Electric Fields and Magnetic Fields behave in our universe
- Show how Electric Fields and Magnetic Fields are intricately linked with each other and perhaps be leveraged to derive wave equations

Those Equations can be used to derive the Wave
Equations
and calculate the speed of light

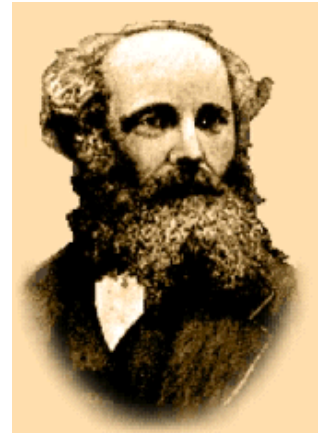
Electromagnetic Wave
Equations

$$\frac{\partial^2 \vec{E}}{\partial t^2} = c^2 \nabla^2 \vec{E}$$

$$\frac{\partial^2 \vec{B}}{\partial t^2} = c^2 \nabla^2 \vec{B}$$

Fun Facts

- When Maxwell worked out his theory of electromagnetism, he ended up with not 4 but 20 equations that describe the behavior of electric and magnetic fields (1861).
- It was Oliver Heaviside in Great Britain and Heinrich Hertz in Germany who combined and simplified Maxwell's Equations into four equations in the two decades after Maxwell's death.



**James Clerk
Maxwell
(1831-1879)**

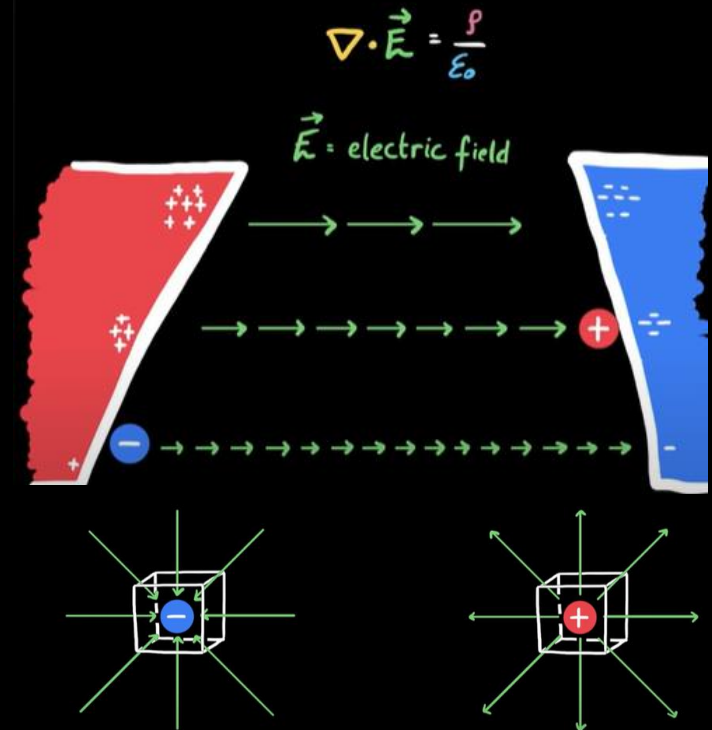
Unpacking Maxwell's Equations

1st Equation: Gauss's Law for Electricity

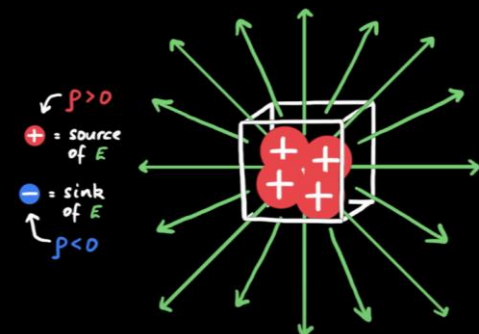
$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Divergence of electric field = $\frac{\text{charge density}}{\text{electric const.}}$

Flux of E through a closed surface = net charge inside the volume



Divergence = field out - field in < 0 Divergence = field out - field in > 0

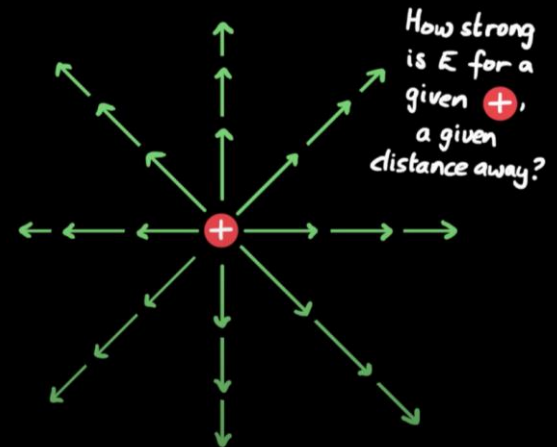


1st Equation: Gauss's Law for Electricity

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Divergence of electric field = $\frac{\text{charge density}}{\text{electric const.}}$

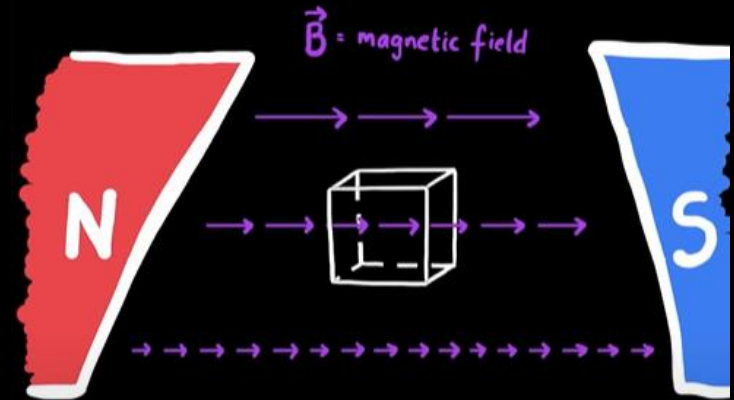
ϵ_0 = permittivity of free space
or electric constant



2nd Equation: Gauss's Law for Magnetism

$$\nabla \cdot \vec{B} = 0$$

Divergence of magnetic field = zero



$$\text{Divergence} = \text{field out} - \text{field in} = 0$$

- Flux of B through a closed surface = 0

Why Electric Field and Magnetic Field behave differently?

- We can't have magnetic monopole

$$\nabla \cdot \vec{B} = 0$$

Divergence of magnetic field = zero

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Divergence of electric field = $\frac{\text{charge density}}{\text{electric const.}}$

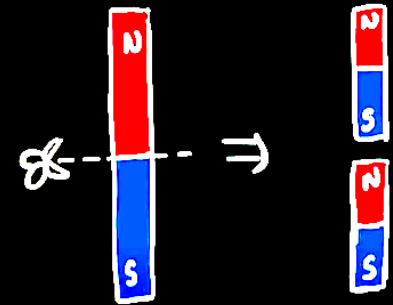
Magnetic field: NO sources or sinks since
 $\vec{\nabla} \cdot \vec{B} = 0$, NOT $\vec{\nabla} \cdot \vec{B} > 0$ $\times$
 or $\vec{\nabla} \cdot \vec{B} < 0$ $\times$

Electric field: CAN have

$$\vec{\nabla} \cdot \vec{E} > 0$$



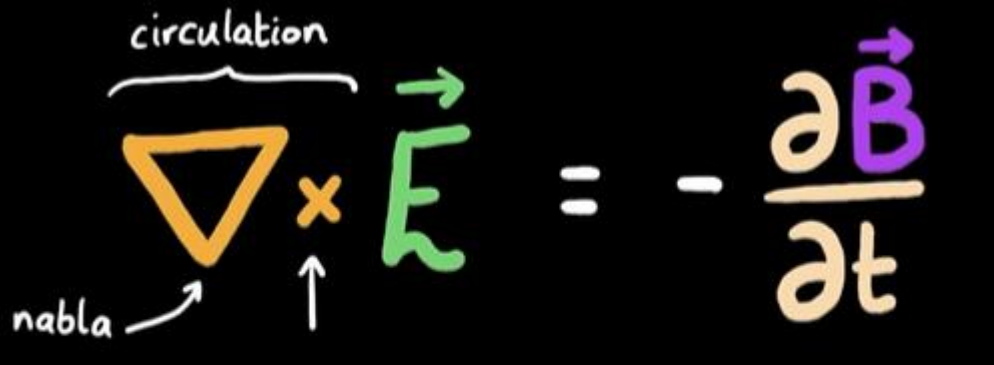
$$\vec{\nabla} \cdot \vec{E} < 0$$



CANNOT
HAVE



3rd Equation: Faraday's Law of Induction


$$\oint \vec{E} \cdot d\vec{l} = - \frac{\partial \vec{B}}{\partial t}$$

curl of electric field = - (rate of change of magnetic field)

- Circulation of E around a closed curve = net change through the surface

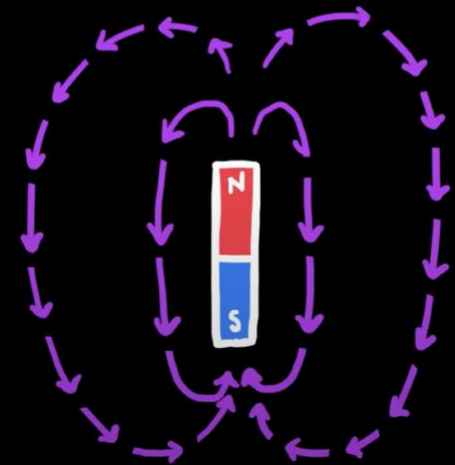
4th Equation: Ampere-Maxwell law

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

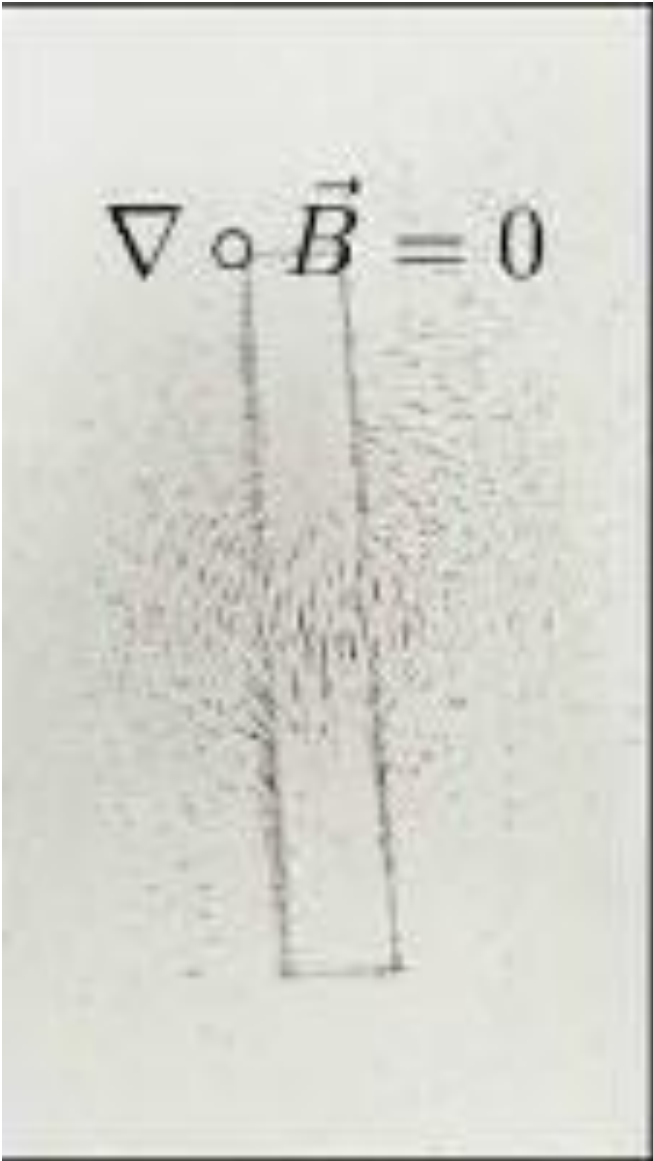
curl of magnetic field = magnetic const. $\left(\text{displacement current} + \text{electric const.} \times \text{rate of change of electric field} \right)$

μ_0 = permeability of free space
or magnetic constant

A circulation of magnetic field can be generated by a changing electric field within our region of space OR a flow of charge (current) IN or OUT of our region



Shorts Era (30 seconds)



$$\begin{aligned}\nabla \cdot \vec{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} &= \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)\end{aligned}$$

Forms of Maxwell's Equation (General Expression or Dynamic or Time-Varying)

Integral Form

Point form

$$\oint_S \vec{E} \circ \hat{n} da = q_{\text{in}}/\epsilon_0 \xrightarrow[\text{theorem}]{\text{Divergence}} \vec{\nabla} \circ \vec{E} = \rho/\epsilon_0 \quad \text{Gauss's law for electric fields}$$

$$\oint_S \vec{B} \circ \hat{n} da = 0 \xrightarrow[\text{theorem}]{\text{Divergence}} \vec{\nabla} \circ \vec{B} = 0 \quad \text{Gauss's law for magnetic fields}$$

$$\oint_C \vec{E} \circ d\vec{l} = -\frac{d}{dt} \int_S \vec{B} \circ \hat{n} da \xrightarrow[\text{theorem}]{\text{Stokes'}} \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{Faraday's law}$$

$$\oint_C \vec{B} \circ d\vec{l} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d}{dt} \int_S \vec{E} \circ \hat{n} da \right) \xrightarrow[\text{theorem}]{\text{Stokes'}} \vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \quad \text{Ampere-Maxwell law}$$

*in a source-free region $\rho = 0$ and $\vec{J} = 0$

** for static $\frac{d}{dt} = 0$

Constitutive Relations

$$\overline{B} = \mu \overline{H}$$

$$\overline{D} = \epsilon \overline{E}$$

(simplification valid **only** if μ and ϵ are constant with respect to frequency)

$$\vec{\nabla} \cdot \overline{D} = q_v \quad \text{OR} \quad \iint \overline{D} \cdot d\vec{S} = \iiint q_v dV$$

$$\vec{\nabla} \cdot \overline{B} = 0 \quad \text{OR} \quad \iint \overline{B} \cdot d\vec{S} = 0$$

$$\vec{\nabla} \times \overline{E} = -\frac{\partial \overline{B}}{\partial t} \quad \oint \overline{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint \overline{B} \cdot d\vec{S}$$

$$\vec{\nabla} \times \overline{H} = \overline{J} + \frac{\partial \overline{D}}{\partial t} \quad \oint \overline{H} \cdot d\vec{\ell} = I + \iint \frac{\partial \overline{D}}{\partial t} \cdot d\vec{S}$$

Summary

- In Maxwell's Equations, you'll encounter two kinds of electric field: the **electrostatic field** produced by electric charge and the **induced electric field** produced by a changing magnetic field.
- Similarly, there are two fundamental types of magnetic fields: **Static (or Permanent) Magnetic Field**: This type of magnetic field is associated with permanent magnets or steady currents and **Induced (or Time-Varying) Magnetic Field**: This type of magnetic field is produced by changing electric fields, according to the amended Ampère's law, which incorporates the concept of displacement current.

Resources

- History of Maxwell's Equations:
https://www.youtube.com/watch?v=3tan4_yh5dw